

Structural properties of gold-silicon nanohybrids formed by femtosecond laser ablation in water at different fluences

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ABSTRACT

A gold target was ablated by femtosecond laser radiation in aqueous solutions of preliminarily prepared Si nanoparticles. The ablation process led to the formation of Au-based spherical colloids with the mean size around 5-10 nm and a weak abundance of larger species. Transmission Electron Microscopy (TEM) and Energy Dispersive X-ray (EDX) analysis revealed the presence of Au and Si in colloid composition, while the stoichiometry of colloids did not depend on laser fluence during the fabrication experiments. The formation of Au-Si nanohybrid structure was explained by an effect of the interaction of laser-ablated Au nanoclusters with water-dispersed Si nanoparticles. The fabricated structures can be of importance for biomedical, catalysis, and photovoltaics applications.

Keywords: hybrid Au-Si nanoparticles, femtosecond laser ablation in water

INTRODUCTION

Silicon nanostructures have attracted a lot of interest since the discovery of visible photoluminescence from porous silicon [1]. These nanostructures have been considered as candidates for optoelectronics [2], photovoltaics [3,4], and various sensing [5-7] applications. Si nanostructures can also offer a large panel of parallel imaging and therapeutic modalities, including bioimaging [8-10], light-induced generation of singlet oxygen and related photodynamic therapy functionality [11-13], infrared radiation [14] radiofrequency radiation [15] ultrasound-induced [16] hyperthermia for cancer treatment. On the other hand, noble metal nanostructures are also considered as very promising candidates for a variety of tasks, including photovoltaics [17], ultrasensitive biological sensing/identification [18-20], Surface Enhanced Raman Scattering [21]. One can imagine that the combination of Au and Si in nanostructure composition could provide new interesting properties for a variety of applications.

It is important that Au and Si nanoparticles (NPs) can be easily prepared by purely physical methods such as pulsed laser ablation in gaseous [22,23] or liquid [24-31] environment. Such an approach is unique in avoiding any residual contamination of nanoparticle surface as a result of purely physical synthesis in pure, well-controlled environment (e.g., deionized water), which looks as a huge advantage over conventional electrochemical [1,2,11,12] or colloidal chemistry [32] routes for the fabrication of colloidal Si or Au-based nanomaterials. Indeed, bare (ligand-free), non-contaminated nanoparticles prepared by laser ablation in liquid ambience have demonstrated unique response in many tasks, including cancer theranostics [33,15], treatment of neurological diseases [34], biofuel cells [35], etc. The nanoparticles can further be functionalized by various biocompatible molecules for appropriate biomedical applications [36].

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In this paper, we explore the possibility of the formation of hybrid silicon-gold nanocolloids by femtosecond laser ablation in deionized water and examine their structural properties. All properties of synthesized hybrid nanostructures are compared with pure Au nanoparticles synthesized under similar laser ablation conditions.

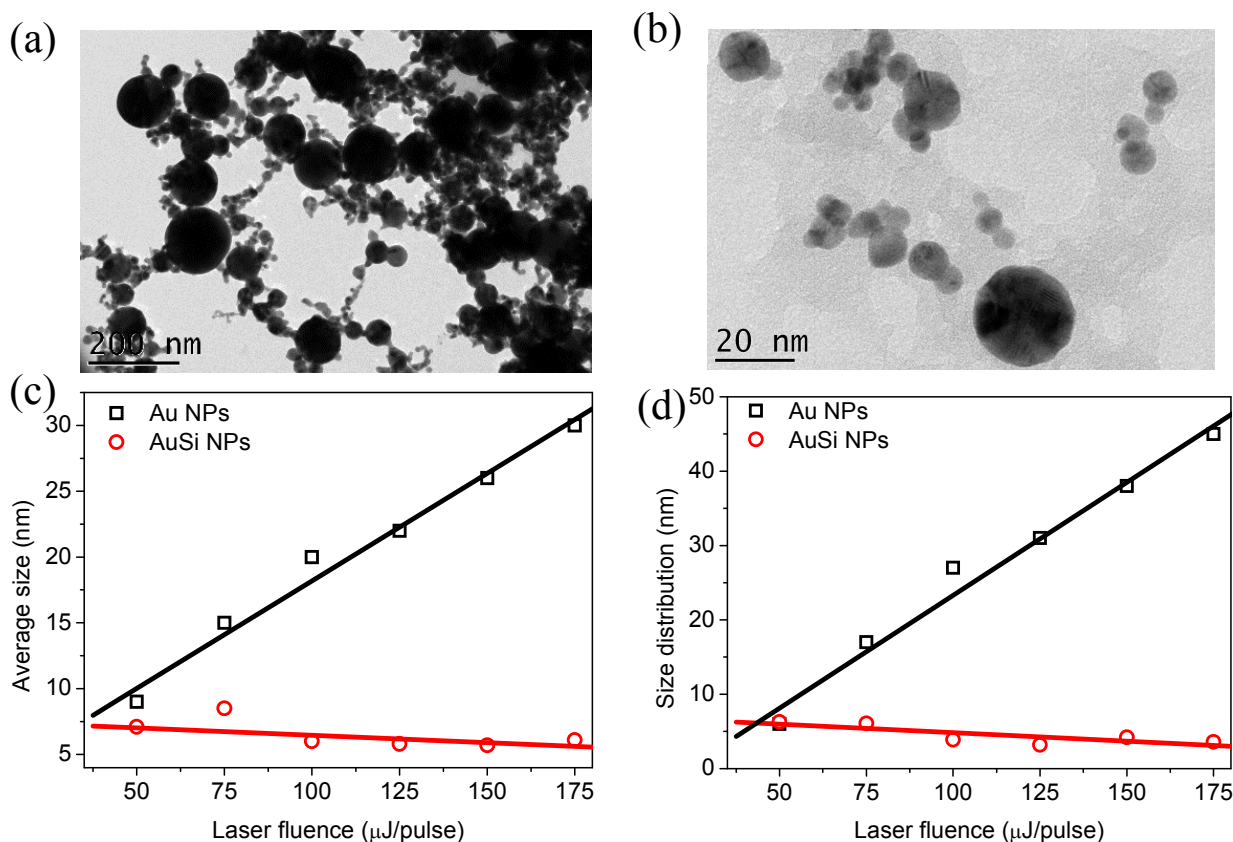


Figure 1. TEM images of bare Au NPs (a) and hybrid AuSi NPs (b) prepared by laser ablation under irradiation with laser energy of 175 $\mu\text{J/pulse}$. Dependence of average size (c) and size dispersion (d) of Au NPs and AuSi NPs on laser energy.

EXPERIMENTAL RESULTS

Laser ablation from a gold target (99.99%) was performed by means of Ti/sapphire laser (800 nm, 110 fs pulse duration, 1 kHz repetition rate). The radiation was focused on the target surface by a 75-mm focal distance lens. A half-wave plate was used for changing laser fluence. The experiments were carried out in pure deionized water or in aqueous solutions of silicon nanoparticles, which were prepared prior to the experiment by methods of laser ablation. In the latter case, a Si target (N-type, (100), 10 $\Omega\cdot\text{cm}$) was ablated by the same femtosecond radiation in deionized water. The mean size of Si nanoparticles was 5-10 nm.

A transmission electron microscope (TEM) JEOL JEM 3010 was used to analyze size distribution and shape of the formed NPs. A droplet of the nanoparticle solution was deposited on a copper grid covered with a carbon film and dried. In order to study chemical composition of hybrid AuSi NPs, we used Energy Dispersive X-ray (EDX) spectroscopy.

Figure 1a shows a typical TEM image of pure Au NPs produced by femtosecond laser ablation from a solid Au target in deionized water. One can see that such a kind of ablation leads to a relatively size-dispersed distribution of nanoparticles with the mean size ranging from a few nm to hundreds of nm. As we earlier showed [28, 29, 37], such a wide dispersion

of nanoparticle sizes is due to radiation-based ablation material and the effect of a cavitation bubble collapse. As shown in Fig. 1b, the ablation from Au target in aqueous solutions of Si nanoparticles led to a quite different distribution of nanoparticles. Here, the mean size of nanoparticles was much smaller (5- 8 nm), although a certain number of larger nanoparticles (10-70 nm) were present (the presence of latter nanoparticles did not affect the final statistics on nanoparticle size distribution). Dependences of the average size of nanoparticles and their size dispersion (full width at half maximum, FWHM) on applied laser fluence is shown in Fig. 1c and Fig. 1d, respectively. Here, one can mention a strong laser energy dependence of size for pure Au NPs. Indeed, the mean size of nanoparticles increased from 8-10 to 30 nm under the increase of laser energy from 50 to 175 $\mu\text{J}/\text{pulse}$. In contrast, the increase of laser energy did not affect the size of nanoparticles under the ablation in solutions of Si nanoparticles, as the mean size and size dispersion were always around 5-8 nm and 3-5 nm FWHM, respectively (Fig. 1c,d). Such a stability of size characteristics of the formed nanoparticles evidences the presence of a strong interaction between Au nanoclusters and Si nanoparticles from the solution.

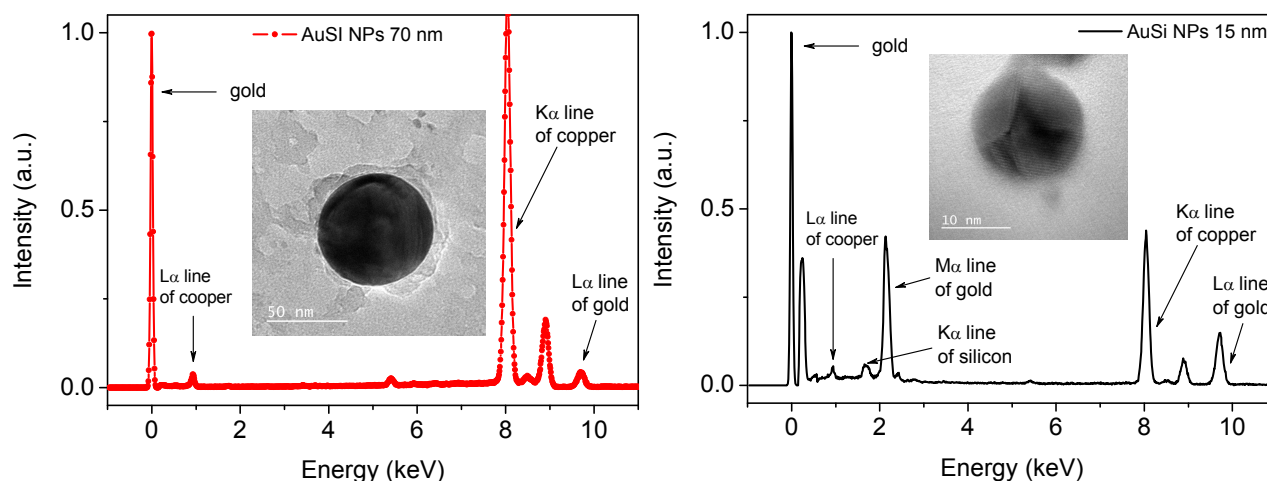


Figure 2. EDX spectra of large (a) and small (b) hybrid AuSi NP

Chemical composition of laser-synthesized NPs was studied for nanoparticles of different sizes using EDX-spectroscopy. Figure 2 shows X-ray responses from the nanoparticles prepared by ablation of Au in deionized water (a) and in aqueous solutions of Si nanoparticles (b). Here, signals at 0.9 keV and 8.0 keV correspond to K_{α} and L_{α} lines of copper, respectively. These signals obviously come from a copper grid used in our experiments. Signals at 2.1 keV and 9.7 keV are related to M_{α} and L_{α} lines of gold, respectively. These signals appear both for nanoparticles synthesized in water and for weakly abundant large nanoparticles (30-100 nm) formed in aqueous solutions of Si nanoparticles, while no EDX signals of silicon were obtained for the latter NPs. However, quite different spectra were obtained for small nanoparticles synthesized under fs laser ablation of Au in aqueous solutions of Si NPs. Here, we could clearly observe K_{α} line of silicon at 1.7 keV, suggesting the presence of Si in nanoparticle composition (7 % for 15 nm AuSi NPs). Thus, we can conclude that the laser ablation procedure really led to the formation of hybrid Au-Si nanoparticles. Finally, we studied chemical composition of selected 15 nm AuSi NPs prepared at different laser fluences. One can see that no dependence of Au/Si ratio of AuSi NPs on used laser fluence was obtained. As shown in Fig. 3, the increase of laser energy did not change the proportion of Si in the nanoparticle composition, which again evidenced the presence of a strong interaction between laser-ablated Au nanoclusters and water-dispersed Si NPs.

DISCUSSION

Let us consider phenomena leading to the formation of Au-Si nanohybrid structures. The action of laser radiation causes ablation of material from the solid target, leading to creation of large amount of gold nanoclusters. Afterwards, they

coalesce creating gold nanoparticles [27]. The increase of the laser fluence can much increase the number of ejected nanoclusters and their energy. When ablated in deionized water, gold nanoclusters do not have any limiting mechanisms and can grow to several tens and even hundreds of nm (Fig. 1c,d). In contrast, when they are ablated in aqueous solutions of Si nanoparticles, they can interact with Si nanoparticles and this process should definitely stop their growth similarly to the case of biopolymers [33] or oligosaccharides [36]. In fact, we observed this phenomenon during our experiments, as the mean size of formed Au-Si nanohybrids was much smaller than pure Au nanoparticles (Fig. 1). Furthermore, the independence of size of AuSi NPs of applied laser fluence indicates a much stronger interaction between gold and silicon species than that of between purely gold nanoclusters. Despite a relatively small mean size of AuSi NPs, some large particles can also be generated and these NPs mainly consisted of pure gold. We can imagine that the generation of these large NPs was due to a casual local drop of concentration of Si nanoparticles. In this case, the concentration of Si nanoparticles was too small to prevent the coalescence of Au nanoclusters. We believe that the increase of Si NPs concentration and a better mixing of solutions should prevent the generation of Au NPs. It is interesting that the increase of laser fluence does not affect the chemical composition and size of AuSi NPs. This confirms our supposition about a very strong interaction of Au nanoclusters and Si nanoparticles.

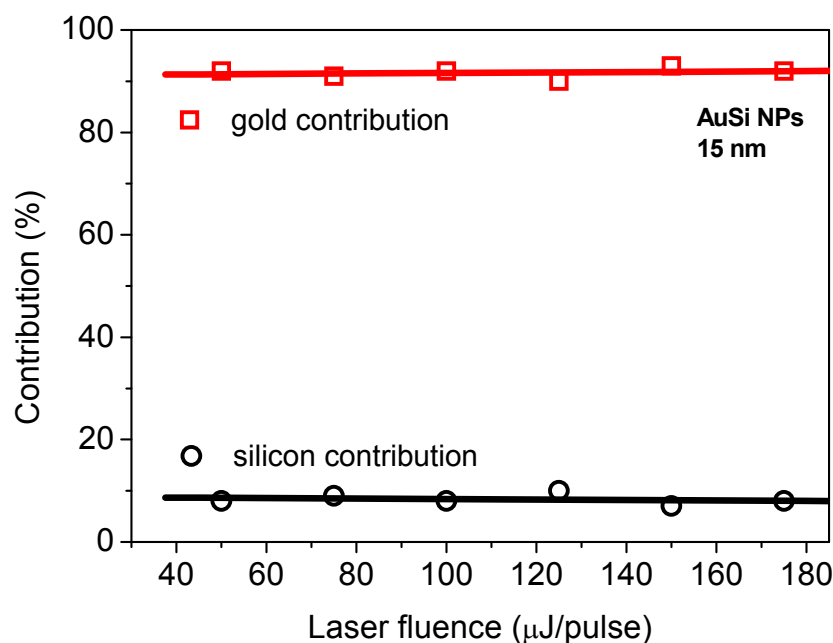


Figure 3. Dependence of gold and silicon contribution of 15 nm AuSi NPs on applied laser fluence.

3. SUMMARY

We studied laser-induced formation of gold-silicon nanoparticle in water at different laser fluences. We managed to synthesize 5-10 nm hybrid Au-Si nanoparticles by fs laser ablation from a gold target in solutions of Si nanoparticles. The mean size, size dispersion and composition of Au-Si nanohybrids are independent of pumping laser energy. The formed nanoparticles can be of interest for biomedical, photovoltaic and optoelectronics applications.

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